

Epidermolytic Hyperkeratosis

The disease is named for its distinctive histopathologic features: **vacuolar degeneration of the epidermis** (epidermal lysis) and **hyperkeratosis**. It is also known as *bullous congenital ichthyosiform erythroderma*, a descriptive term reflecting the blistering presentation at birth, scaling, and erythema.

Inheritance and Epidemiology

Epidermolytic hyperkeratosis is inherited as an **autosomal dominant** trait, with a prevalence of approximately **1 in 200,000 to 300,000 individuals**. However, **spontaneous mutations** are common, and up to **half of cases** lack a family history, representing new mutational events.

Clinical Presentation

The disease typically presents **at birth** with: - Blistering - Erythema (redness) - Peeling

Over time, **generalized hyperkeratosis** develops, which may persist with or without erythroderma. A **characteristic pungent odor** is often present, likely due to superinfection by mixed bacterial flora.

Histopathology

Unlike most other ichthyoses, epidermolytic hyperkeratosis has a **distinctive histologic appearance**: - Markedly **thickened stratum corneum** - **Vacuolar degeneration** of the upper epidermis (epidermolysis) - Involvement typically affects the upper epidermis, sometimes all suprabasal keratinocytes - **Granular cells** contain dense, enlarged, irregular keratohyalin granules - **Electron microscopy** reveals clumping of keratin intermediate filaments (keratin-1 and keratin-10) beginning in the first suprabasal layer

Clinical Heterogeneity and Subtypes

There is significant clinical variation among affected individuals. Six clinical phenotypes have been described, categorized based on: - Presence or absence of **severe palmar/plantar hyperkeratosis** - Presence of **erythroderma** - Scale characteristics - Extent of skin involvement - Digital contractures - Gait abnormalities

NPS Types (No Severe Palmar/Plantar Hyperkeratosis)

- **NPS-1:** Generalized thick, brown, verrucous hyperkeratosis, most prominent over joints. Palms and soles are spared. Characteristic **cobblestone or hystrix (porcupine-like) spiny papules**. This is a severe and commonly recognized form.
- **NPS-2:** Milder than NPS-1; lacks hystrix spines and shows relative sparing of inter-joint skin.
- **NPS-3:** Presents with **generalized exfoliative erythroderma**.

PS Types (With Severe Palmar/Plantar Hyperkeratosis)

- **PS-1:** Smooth palmar/plantar hyperkeratosis with sharp borders, often surrounded by a red halo. Blisters may occur at the border. Trunk involvement is limited, usually over joints.
- **PS-2:** Generalized erythroderma and scaling. Severe volar involvement may lead to **digital contractures** and **ainhum-like constricting bands**.
- **PS-3:** Generalized pebbly hyperkeratosis with a **cerebriform (brain-like) pattern** on palms and soles.

Mosaic (Sporadic) Form

Sporadic cases due to **postzygotic mutations** during embryogenesis can present as a **mosaic pattern**: - Streaks or patches of hyperkeratotic skin alternating with normal skin - Distribution follows **Blaschko's lines** - May be limited or widespread - Can be **unilateral** in localization